

Software Engineering Lab-1

Requirements engineering & UML use-case modelling

Name – Sreyaesh Peruri

SRN – PES1UG24AM189

Scenario number - 11

Telemedicine Slot booking & Prescription Portal

1. Functional Requirements Table

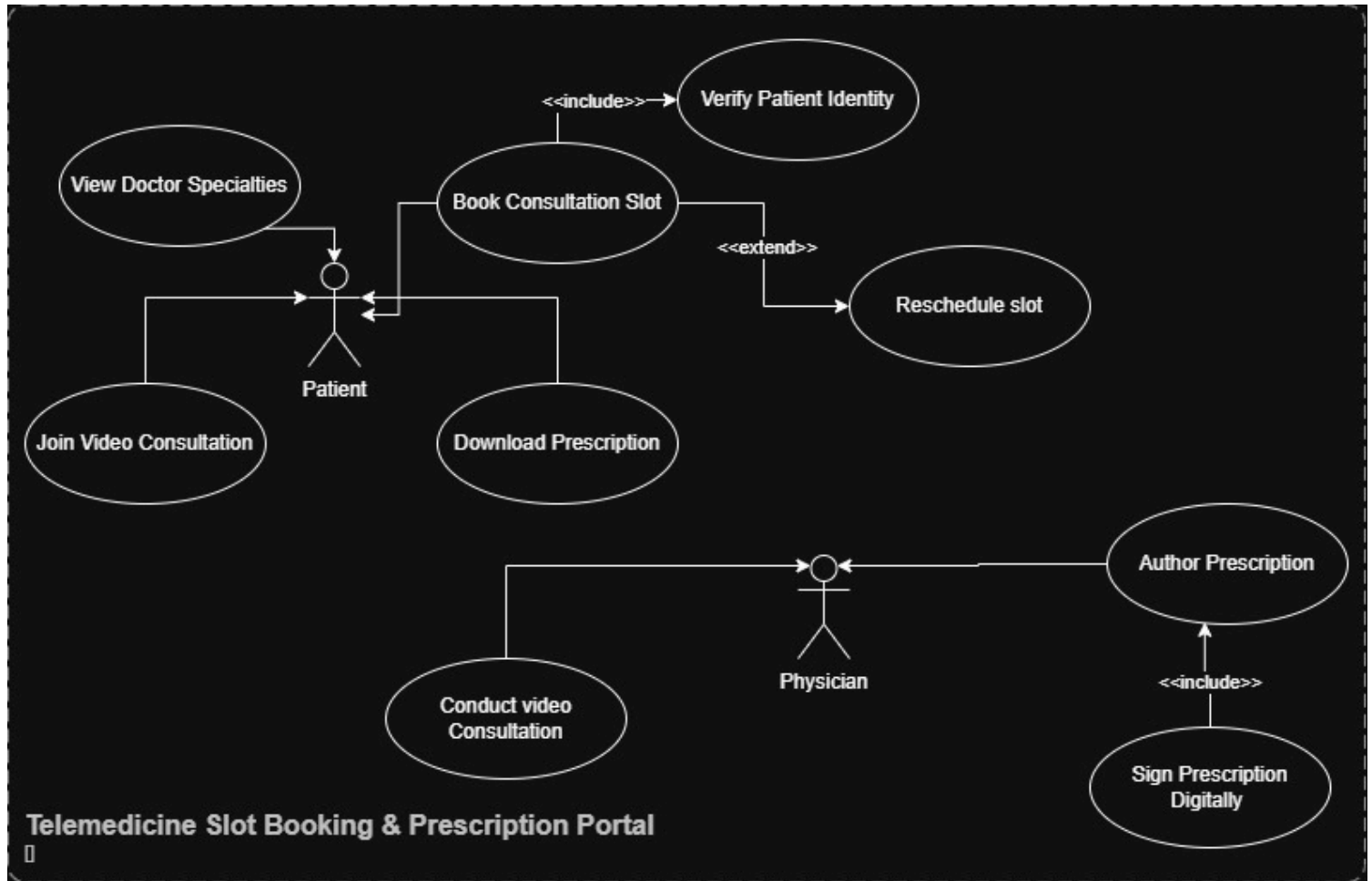
ID	Description	Priority	Acceptance Criteria
FR-001	The system shall allow patients to view a list of doctors filtered by specialty and available consultation time slots.	High	Pass: Only doctors matching the selected specialty and currently open slots are displayed. Fail: Booked or past slots appear as available.
FR-002	The system shall allow patients to book a video consultation slot with a selected physician for a specific date and time	High	Pass: Slot is reserved and marked unavailable to other patients immediately after confirmation. Fail: Two patients are able to book the same slot simultaneously.
FR-003	The system shall automatically generate and send an encrypted, unique video consultation room link to both patient and physician at least 15 minutes before the scheduled appointment.	High	Pass: Both parties receive a valid, working encrypted link before the session start time. Fail: Link is reused across sessions or accessible without authentication.
FR-004	The system shall enable physicians to author and digitally sign structured	High	Pass: Valid cryptographic signature is attached to the

	prescriptions that are automatically compiled into a downloadable PDF record.		prescription PDF. Fail: Prescription is generated without doctor license/identity validation.
FR-005	The system shall allow patients to securely download their signed prescription PDF from their account after consultation.	Medium	Pass: Only the patient associated with the consultation can access and download the file. Fail: Prescription is accessible without patient authentication.

2. Non - Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Relationale
NFR-001	Performance & Security	All video session metadata and patient clinical notes must be encrypted in transit using TLS 1.3 and at rest using AES-256.	High	Pass: Benchmarking confirms encryption standards are met under simulated peak load. Fail: Any unencrypted data transmission or storage is detected	Protects sensitive patient health data (PHI) from interception or breach, ensuring regulatory compliance
NFR-002	Usability & Availability	The video consultation platform shall maintain 99.5% uptime and load the booking interface within 3 seconds on standard broadband connections.	Medium	Pass: System uptime and load-time benchmarks are met over a 30-day monitoring period. Fail: Downtime exceeds 0.5% or interface load time exceeds 3 seconds.	Ensures reliable access for patients seeking timely medical care, minimizing missed appointments due to platform failure.

3. UML Use-case Diagram



4. Use-case Flow Specification

1. Use Case Name: Book Consultation Slot

2. Actor: Patient

3. Preconditions:

- Patient is registered and logged into the system
- Patient's identity has been verified
- At least one doctor has open available slots

4. Postconditions:

- Slot is reserved and marked unavailable to other patients

- Booking confirmation is sent to the patient
- Consultation appears in both patient's and physician's schedules

5. Main Success Scenario:

1. The patient logs into the portal.
2. Patient searches/browses doctors by specialty.
3. The system displays a list of doctors with available time slots.
4. The patient selects a doctor and an available time slot.
5. System verifies patient identity (<<include>> Verify Patient Identity).
6. The patient confirms the booking.
7. The system reserves the slot and marks it unavailable to others.
8. The system sends booking confirmation with appointment details to patient and physician.

6. Alternate Flow:

A1: Slot becomes unavailable during booking

- At step 6, if another patient books the same slot first, the system detects the conflict.
- The system notifies the patient the slot is no longer available.
- The system returns the patient to the doctor's slot list to choose a different time.
- Flow resumes at step 4.